Mobile App Update Impact and Failure Detection

Final Project Report (Units 1–3)

Course: 23CSE452 Business Analytics

Review 2: Advanced Analytics (Unit 3): Text Mining and Time-Series Analysis

Team number: 10

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Link	
Live dashboard	https://team10-app-reviews.streamlit.app/
Repository	https://github.com/Adithya-Monish-Kumar-K/ba-capstone-app-reviews
Project board	https://github.com/users/Adithya-Monish-Kumar-K/projects/4
Final notebook	`notebooks/Review_2.ipynb` (executed; HTML copy `notebooks/Review_2.html`)

Summary

We collected every Play Store review of 11 Indian consumer apps from 1 April to 20 September 2026 (1,137,987 reviews after cleaning). Review 1 cleaned the data, tagged 9 issue types, scored sentiment and built a classifier that flags problem reviews. Review 2 applies two Unit 3 methods to the same data and combines everything into recommendations and a live dashboard.

- **Method 1, text mining.** NMF topic models on 167,303 complaint reviews find 17 complaint topics across the three domains; on held-out reviews NMF is more coherent and more distinct than LDA in every domain. The topics agree with the keyword tags where they should, and they also cover the 43% of complaints (71,775) that match no keyword tag. Refund and cancellation problems lead Food & Grocery (35% of complaints), returns, refunds and product quality lead Shopping (25%), and payment, UPI and bank-account problems lead Payments (49%). Among specific faults, poor support draws the angriest reviews (mean VADER −0.40 in food apps); technical faults are reported calmly (−0.09).
- **Method 2, time series.** ARIMA/SARIMA models forecast the daily number of 1–2★ reviews per app. On a rolling-origin backtest of 252 forecasts per app, with every modelling choice made on the training days only, ARIMA beats the seasonal-naive baseline for 10 of 11 apps (mean MASE 0.93 against 1.08) and is as accurate as Holt-Winters; it is kept for its prediction intervals. Its 80% and 95% intervals contain 83% and 94% of the days, and no app has autocorrelation left in its residuals. The forecast flags 77 unusual days. Of 205 testable app releases, 196 changed nothing measurable, 5 made things better and 4 made things worse. Only one, Paytm 10.84.5, is confirmed at the app level: Paytm's 1–2★ reviews per day went from 34 to 66.
- **Combined insights.** Six warning signs, drawn from every stage, give each app a priority. Swiggy, Flipkart, Amazon and Paytm are High priority. The signs also produce 65 recommendations, each backed by a number.

- **Dashboard.** One Streamlit app with six pages, deployed publicly. It covers the overview, text mining, forecasting, update impact, recommendations and checking a single review.

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1. Problem, Questions and Methods

1.1 Problem

Apps like Swiggy, Zomato, Blinkit, Flipkart, Amazon, Paytm and PhonePe get hundreds of Play Store reviews a day, and the busiest get more than a thousand. Many reviews report real problems: a failed payment, a late delivery, a refund that never came, an app that misbehaves after an update. A product or support team cannot read them all, so problems are noticed late.

Review 1 answered "which reviews are problems?" with a classifier: Random Forest PR-AUC 0.87 against a base rate of 0.19. Review 2 asks the questions a manager needs answered next:

1. **What exactly are users complaining about, in their own words, and how angry are they?** Keyword tags miss reviews that are worded differently.
2. **How many complaints will each app get in the coming weeks, which days are unusual, and did a new app version make things worse?**
3. **Which app needs attention first, and what should it do?**

1.2 Methods chosen

The rubric asks for two of the three Unit 3 methods. We chose the two that fit review data best:

Method	Why it fits	Answers
Text mining	The reviews are unstructured text; topic models find themes without a fixed keyword list, and sentiment measures how strongly users feel	Question 1
Time-series analysis	The reviews are dated, so complaints form a daily series per app, and app versions give dated release events	Question 2

Association rule mining was not chosen. Only 3.4% of reviews carry two or more of the 9 issue tags (median review length is 2 words), so there are few co-occurring "items" to mine. Review 1's EDA had already measured how the tags co-occur (lift, EDA chart 7), so rules would repeat that rather than answer a new question.

1.3 How this addresses the research gap

Review 1 (§1.3 of `REPORT.md`) found four gaps in published app-review mining: reviews are studied as one fixed set; labels are generic; tasks are done alone; and collection bias is rarely checked. Review 2 addresses each of them:

- It **follows complaints over time**: daily series, forecasts and before/after tests around 224 releases.
- It finds **app-specific themes** from the text itself: 17 topics, plus the words that set each app apart.
- It **combines** text mining, forecasting, update impact and the classifier into one priority table and one dashboard.
- It carries the **feed gap** found in Review 1 into the models as an input instead of reading it as a real change.

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2. Data

The data is the Review 1 dataset, collected by us from the Google Play Store (see `DATA_SOURCES.md`). It holds 1,137,987 English reviews from India for 11 apps in three domains, covering 1 April to 20 September 2026 (173

days):

Domain	Apps
Food & Grocery	Swiggy, Zomato, Blinkit, Domino's
Shopping	Myntra, Flipkart, Amazon, Meesho
Payments	Paytm, PhonePe, Google Pay

Review 2 uses the following outputs of Review 1:

- the star rating, used to define complaints (1–2★) and praise (4–5★);
- the review text and date;
- the app version on the reviewer's phone;
- the 9 keyword issue tags;
- the VADER sentiment scores;
- the saved classifiers.

There are two known data-quality points:

- **Feed gap.** From 21 April to 5 May, the Play Store feed is missing most positive reviews for Swiggy, Blinkit, Domino's, Flipkart and Amazon.
- **Zomato outage.** Zomato has no reviews from 23 July 22:00 to 25 July 12:30.

2.1 Dataset documentation and validation

Item	File
Final validated dataset	`data/tagged/<app>.csv.gz` (11 files, 1,137,987 rows × 29 variables)
Dataset description and collection method	`DATA_SOURCES.md`
Data dictionary (every variable: type, range, missing share, meaning)	`DATA_DICTIONARY.md`
Sample dataset (10 random reviews per app)	`data/sample/final_dataset_sample.csv`
Validation summary	`DATA_VALIDATION.md`, `data/dataset_validation.json`

`scripts/24_dataset_documentation.py` re-checks the final dataset and regenerates these files. All 18 checks pass:

- The row count matches the cleaning log, and the cleaned and tagged files hold the same reviews.
- There are no duplicate reviews, and no missing values outside `app_version` (13.4% empty, kept, not a model input).
- Ratings are 1–5, every date lies in the window, and every review contains letters.
- Every derived field agrees with its source: month, word count, issue counts, sentiment labels.
- VADER scores are in range. The 47 reviews made only of one-letter tokens ("o k") score 0 on all four, which is how VADER works.

Step	Reviews
Raw reviews collected	1,202,729
Empty or under 3 characters removed	47,533
No letters (emoji or punctuation only) removed	7,420
Non-English removed	9,789

Step	Reviews
Final validated reviews (94.6% retained)	**1,137,987**

2.2 Units 1 and 2 in brief

- **Unit 1 (preprocessing and EDA).** Cleaning, 9 keyword issue tags and VADER sentiment, then 15 EDA charts with bias checks (`EDA.md`). Apps differ widely even within a domain: 54% of Amazon's reviews are 1–2★ against 10% for Myntra. 91% of the 1–2★ reviews that name an issue describe a service problem, not an app fault.
- **Unit 2 (predictive model).** Target: `is_problematic` = 1–2★ (19.5%). Inputs: 200 SVD components of TF-IDF text features plus 15 structural features. Logistic Regression and Random Forest were evaluated on 227,598 held-out reviews (stratified 80/20 split; `MODEL_EVALUATION.md`).

Review 1 fitted TF-IDF, SVD and the scaler on all rows before the split. `scripts/25_train_only_preprocessing_check.py` refits them on the 910,389 training reviews only and changes no metric by more than 0.003 (`data/model_train_only_preprocessing.csv`):

Model (preprocessing fitted on training rows only)	Precision	Recall	F1	PR-AUC
Majority-class baseline	0.000	0.000	0.000	0.195
Logistic Regression	0.770	0.858	0.811	0.871
Random Forest	0.736	0.870	0.798	0.874

3. Method 1: Text Mining

Code: `scripts/16_text_preprocessing.py`, `17_topic_modelling.py`, `18_topic_evaluation.py`, `20_topic_examples.py`. The full write-up is `TEXT_MINING.md`.

3.1 Data preparation

Two corpora. A review is kept if it has at least 3 content words after cleaning, because one-word reviews ("worst", "good") cannot carry a topic.

Corpus	Reviews	Kept	Median kept length
Complaints (1–2★)	221,581	167,303 (75.5%)	11 words
Praise (4–5★), as a contrast	871,380	166,635 (19.1%)	4 words

Cleaning steps:

1. Lowercase the text and remove links, e-mail addresses, emojis and digits.
2. Remove two-word brand names as phrases ("google pay", "phone pe", "phone pay"), so that "pay" and "google" stay usable on their own. ("phone pay", a common spelling of PhonePe, otherwise forms a topic of its own.)
3. Expand contractions, with or without the apostrophe ("didnt" → "did not").
4. Drop stop words but **keep negations**, so that "not delivered" keeps its meaning.
5. Map Hinglish negations ("nahi", "nhi") to "not", and drop Hinglish filler words.
6. Drop app names, so that topics describe problems rather than apps.
7. Lemmatise with WordNet ("refunded" → "refund").

For example, "PLEASE STOP CHEATING US BY CHARGING DELIVERY CHARGES FOR ORDERS ABOVE 100…" becomes `stop cheat charge delivery charge order people number fraud`.

Document-term matrices. There is one matrix per corpus and domain, holding words and two-word phrases that appear in at least 10 reviews and at most half of them:

Matrix	Reviews	Terms
Complaints · Food & Grocery	82,602	17,447
Complaints · Shopping	71,412	18,664
Complaints · Payments	13,289	2,974
Praise · Food & Grocery	74,647	6,876
Praise · Shopping	75,671	7,097
Praise · Payments	16,317	1,745

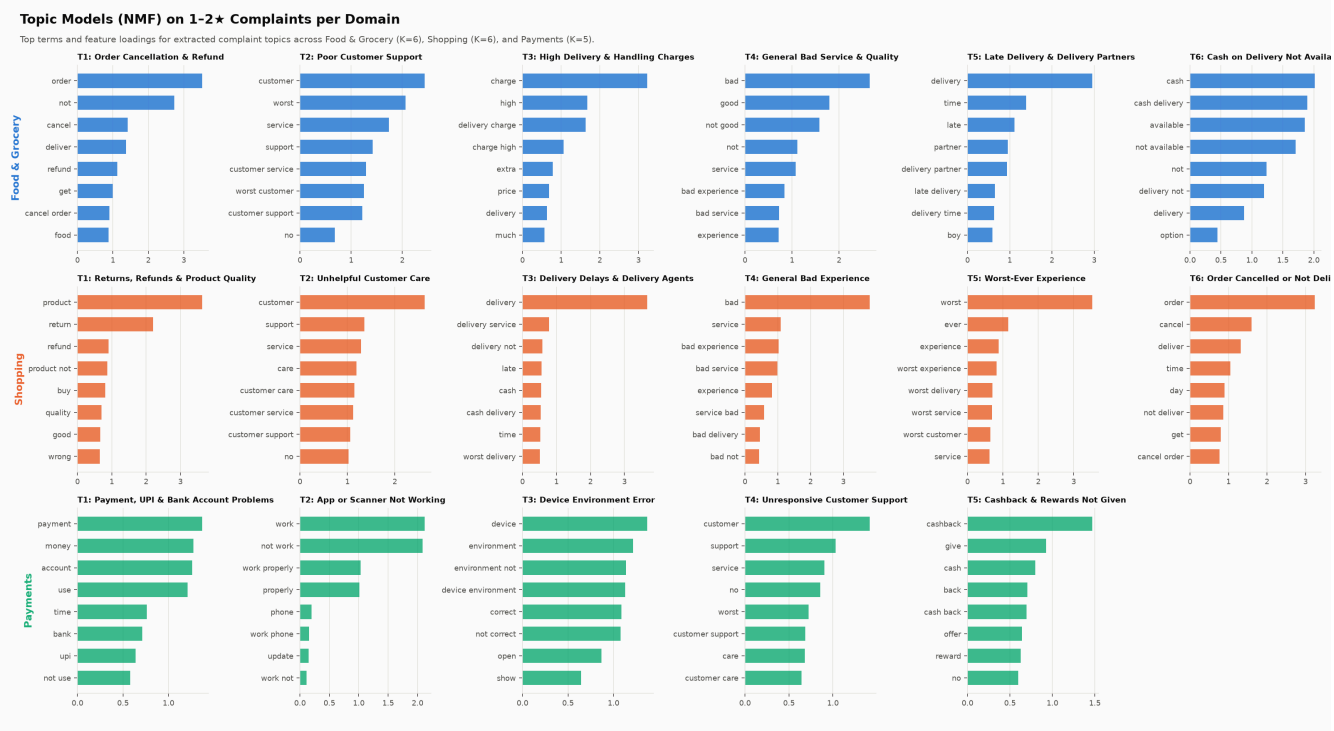
3.2 Implementation: NMF topic models

- **Model.** The counts are converted to TF-IDF weights. Non-negative Matrix Factorisation splits each domain's matrix into document-topic weights and topic-term weights ($V \approx W \times H$) [6].
- **Why NMF and not LDA.** NMF with the `nndsvda` start is deterministic and fits 167,303 reviews in seconds; §3.3 compares it with LDA on held-out reviews.
- **One model per domain.** The models use K = 6 for Food & Grocery, 6 for Shopping and 5 for Payments (chosen in §3.3).
- **Topic and sentiment for every review.** Every review gets its dominant topic. Its VADER compound score then gives each topic a mean sentiment and a share of strongly negative reviews (compound ≤ −0.5). The 105 reviews with no vocabulary term have zero weight on every topic and are marked Unassigned.
- **Topic names** describe each topic's top terms. Three topics are general dissatisfaction ("bad experience", "worst ever") rather than one fault, and are named so.
- **Praise contrast.** A second NMF model (K = 4 per domain) on the 4–5★ reviews shows what satisfied users mention.
- **Saved pipelines.** Each fitted model is saved with its vectoriser in `models/textmining/`, so the dashboard can give a new review its topic.

The 17 complaint topics (`data/textmining/topic_sentiment.csv`):

Domain	Topic	Reviews	Share of domain	Mean VADER	Strongly negative
Food & Grocery	Order Cancellation & Refund	28,653	35%	−0.27	36%
	Poor Customer Support	15,641	19%	−0.40	56%
	Late Delivery & Delivery Partners	14,861	18%	−0.29	39%
	General Bad Service & Quality	10,394	13%	−0.44	59%
	High Delivery & Handling Charges	8,724	11%	−0.21	24%
	Cash on Delivery Not Available	4,304	5%	−0.12	18%
Shopping	Returns, Refunds & Product Quality	18,123	25%	−0.23	33%
	Order Cancelled or Not Delivered	15,812	22%	−0.31	41%
	Unhelpful Customer Care	13,120	18%	−0.24	41%
	Delivery Delays & Delivery Agents	11,150	16%	−0.29	41%
	Worst-Ever Experience	6,976	10%	−0.61	80%

Domain	Topic	Reviews	Share of domain	Mean VADER	Strongly negative
	General Bad Experience	6,195	9%	−0.59	81%
Payments	Payment, UPI & Bank Account Problems	6,443	49%	−0.15	25%
	Unresponsive Customer Support	2,410	18%	−0.23	41%
	Cashback & Rewards Not Given	1,941	15%	−0.15	28%
	App or Scanner Not Working	1,305	10%	−0.09	16%
	Device Environment Error	1,146	9%	−0.13	19%



3.3 Evaluation

Number of topics (coherence). NPMI coherence [7] was computed for K = 3 to 10 in each domain (`coherence_scores.csv`):

- **Food & Grocery.** Coherence peaks at K = 6 (0.195), the value chosen.
- **Payments.** Coherence peaks at K = 5 (0.242), the value chosen.
- **Shopping.** The curve is almost flat from K = 3 to 8 (0.156–0.168); the highest values sit at the edge of the range (K = 9–10). We took K = 6, at which every topic is distinct and can be named.

Coherence decides K for Food & Grocery and Payments; for Shopping it is a guide.

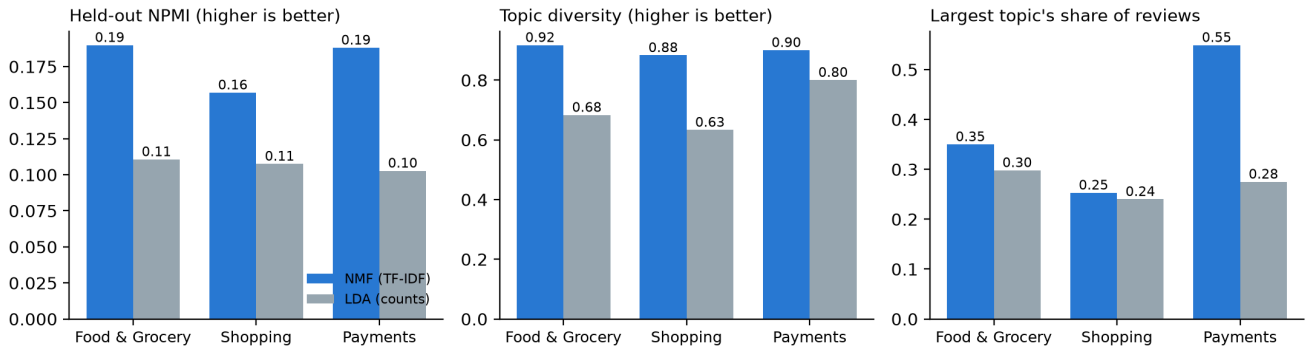
NMF compared with LDA (`scripts/23_topic_model_comparison.py`, `data/textmining/topic_model_comparison.csv`). Both models were given the same document-term matrix and the same K for each domain. Each was fitted on 80% of the domain's complaint reviews (133,841 in total), and its topics were scored on the unseen 20% (33,462):

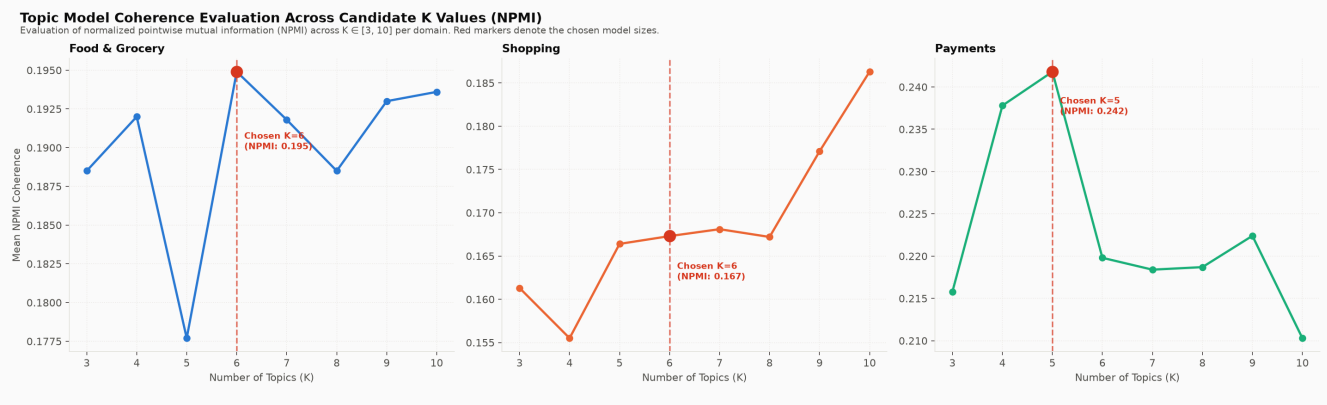
- **Held-out NPMI:** how often each topic's top 10 terms appear together in unseen reviews.
- **Topic diversity:** the share of distinct terms among all topics' top 10 terms.

Domain	Model	Held-out NPMI ↑	Topic diversity ↑	Largest topic's share of reviews	Fit time
Food & Grocery	**NMF (TF-IDF)**	**0.190**	**0.92**	35%	2 s
	LDA (counts)	0.111	0.68	30%	41 s
Shopping	**NMF (TF-IDF)**	**0.157**	**0.88**	25%	1 s
	LDA (counts)	0.108	0.63	24%	41 s
Payments	**NMF (TF-IDF)**	**0.188**	**0.90**	55%	0.1 s
	LDA (counts)	0.103	0.80	28%	7 s

NMF's topics are more coherent and more distinct in every domain, which confirms the choice. LDA spreads reviews more evenly across topics, but its topics share many of the same top terms. K was chosen for NMF and LDA used default settings, so the comparison favours NMF slightly; the gap in held-out NPMI (0.05–0.09) is large compared with the differences between values of K.

NMF vs LDA on the same complaint reviews, same K, scored on a 20% held-out split





Agreement with the 9 keyword tags. We cross-tabulated each topic with the keyword tags from Review 1 (`topic_keyword_overlap.csv`). Topics and tags agree where they should:

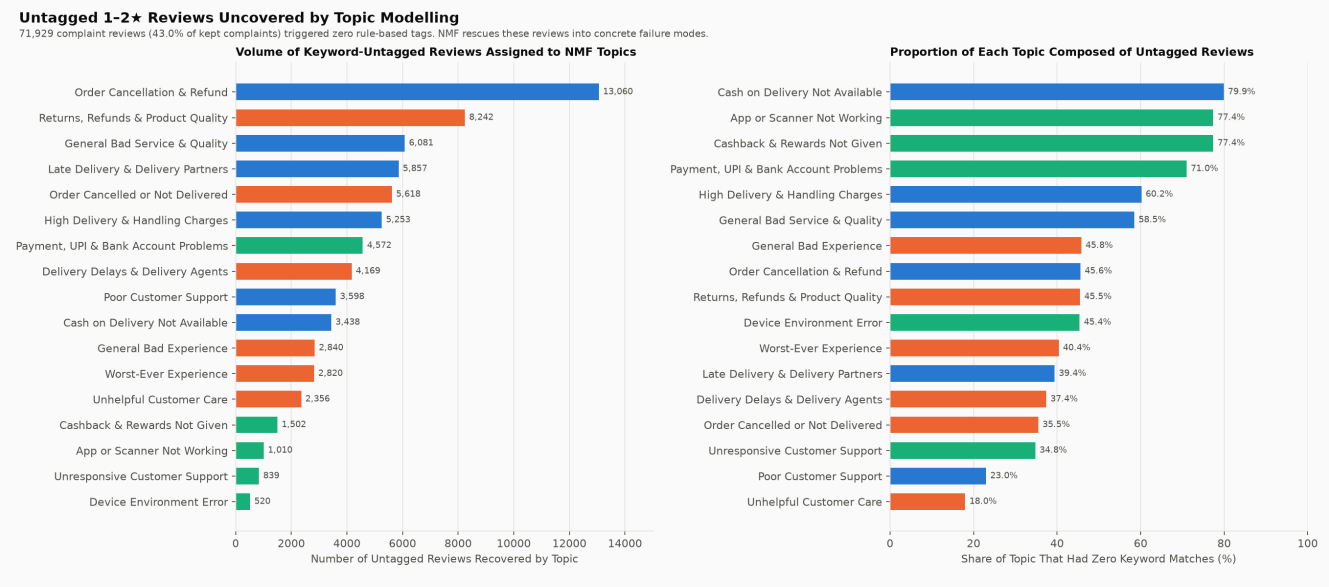
Topic	Matching keyword tag	Share of the topic's reviews with that tag
Poor Customer Support (food)	Customer Support	71%
Unhelpful Customer Care (shopping)	Customer Support	76%
Late Delivery & Delivery Partners (food)	Delivery Delay	51%
Device Environment Error (payments)	Crash & Stability	50%
Returns, Refunds & Product Quality; Order Cancelled or Not Delivered	Cancellation & Return	31–39%

The topics also find themes that no keyword category covers: cash on delivery not being available (80% of its reviews untagged) and cashback or rewards not given (77% untagged).

What the keyword rules miss. 71,775 of the 167,198 complaints with a topic (42.9%) match none of the 9 keyword tags. The topic models give every one of them a theme:

Topic	Untagged reviews	Share of the topic that is untagged
Order Cancellation & Refund	13,060	46%
Returns, Refunds & Product Quality	8,242	45%
General Bad Service & Quality	6,081	59%
Late Delivery & Delivery Partners	5,857	39%
Order Cancelled or Not Delivered	5,618	36%

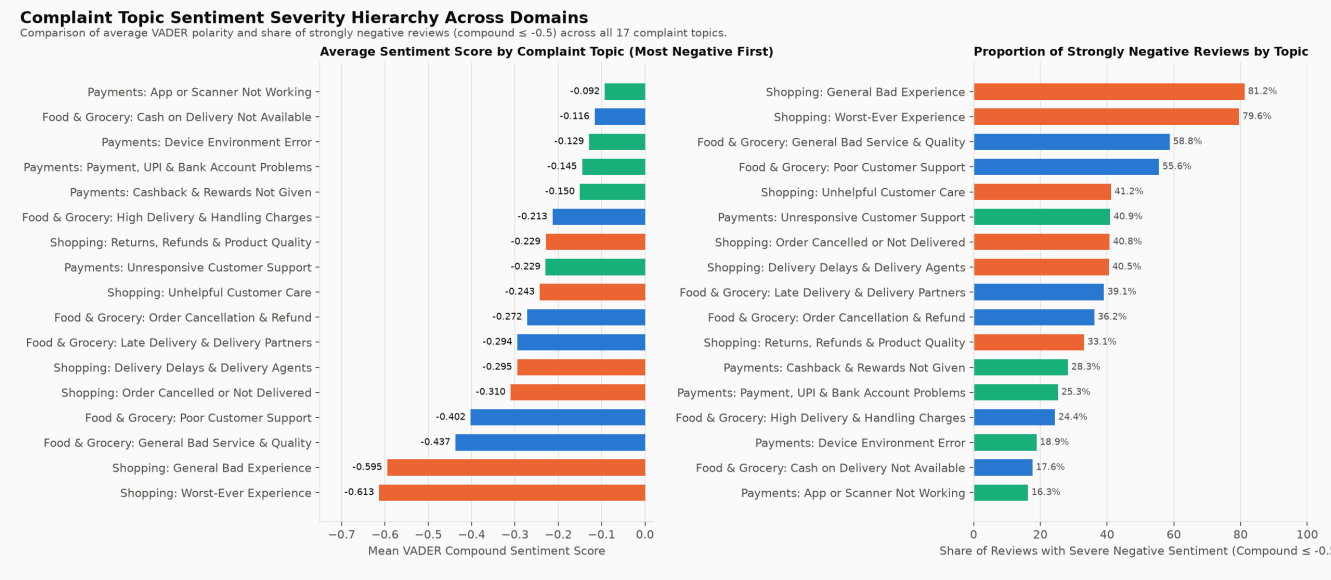
The untagged share is highest for Cash on Delivery Not Available (80%), Cashback & Rewards Not Given and App or Scanner Not Working (77% each) and Payment, UPI & Bank Account Problems (71%): their wording is not in the keyword lists.



3.4 Interpretation

Sentiment by topic.

- **The most negative topics are the general ones.** Worst-Ever Experience (−0.61, 80% strongly negative) and General Bad Experience (−0.59) in Shopping, and General Bad Service & Quality (−0.44) in food apps, are made of words like "worst" and "bad". They measure anger, not a fault to fix.
- **Among specific faults, support draws the most anger.** Poor Customer Support in food apps (−0.40, 56% strongly negative) and Unresponsive Customer Support in payment apps (−0.23) are the angriest specific topics; in Shopping it is Order Cancelled or Not Delivered (−0.31).
- **Technical faults are reported calmly.** In Payments, app or scanner failures (−0.09) and device-environment errors (−0.13) are described factually ("app not working after update"). A mild VADER score is therefore not a sign of a mild problem.
- **In Payments, the anger is aimed at support.** Unanswered support (−0.23) draws more anger than the payment problem itself (−0.15). Cashback and rewards that were promised but not given form their own topic (15% of payment complaints).



How apps differ (distinctive words). Standardised log-odds with an informative Dirichlet prior [8] compare each app with the other apps of its domain:

App	Most distinctive complaint words	Reading
Swiggy	cancellation, cancellation fee, hour	Cancellation fees and long delays
Zomato	restaurant, support, gold, platform fee	Zomato Gold changes and platform fees
Blinkit	product, return, item, damage	Damaged or missing grocery items
Domino's	pizza, store, outlet, cheese	Store-level order problems
Myntra	exchange, tag, size	Size exchanges, and returns refused over tags
Flipkart	minute, fee, delay, reschedule	Flipkart Minutes delays, rescheduled deliveries, fees
Amazon	prime, membership, device, refund	Prime membership and refund disputes
Meesho	account, verification, block	Account verification and blocking
Paytm	device, error, correct	"Device environment not correct" security errors
PhonePe	charge, wallet, fee, block	Charges and fees, the wallet, blocked accounts
Google Pay	reward, pocket, bug	Rewards and app bugs

Praise as a contrast. Satisfied users mention fast delivery (food), product quality and price (shopping), and easy UPI payments (payments).

Topic share over time. The topic mix is stable from April to September:

- Order Cancellation & Refund stays at 34–36% of food complaints.
- Customer support rises slightly in Food & Grocery (18% to 20%) and in Payments (18% to 20%).
- Payment, UPI and bank-account problems move between 46% and 52% of payment complaints.

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4. Method 2: Time-Series Analysis

Code: `scripts/15_time_series_data.py`, `19_forecasting.py`, `20_forecast_evaluation.py`, `21_update_impact.py`. The full write-up is `TIME_SERIES.md`.

4.1 Data preparation and patterns

The series. The modelled series is the daily count of 1–2★ reviews per app: 11 series of 173 days. We use counts rather than shares because the feed gap removed mostly positive reviews, which distorts shares far more than counts.

Data-quality flags:

- **Feed gap.** The feed-gap days (21 April – 5 May, five apps, 75 app-days) become a 0/1 input to the model.
- **Zomato outage.** Zomato's one fully missing day (24 July) is filled with the mean of the same weekday one and two weeks either side. The two partial days keep their real counts.

Release events. The first day an app version appears in the reviews marks its rollout. Of 2,507 versions, 224 count as new releases. A version counts if it:

- has at least 30 reviews;
- first appears after 1 April; and
- is numbered higher than every earlier version, which removes old builds that some users still run.

Patterns.

- **Trend (STL decomposition [9]).** Paytm rises the most (+60%, first four weeks against the last four), followed by Myntra (+22%), Amazon (+14%) and Blinkit (+13%). Google Pay, Domino's, Swiggy and Zomato fall.
- **Weekly pattern.** Food apps get 22–38% more complaints on Sundays: Domino's +38%, Swiggy +28%, Zomato +28% and Blinkit +22%. Shopping apps have no weekday pattern.
- **Stationarity.** The ADF test rejects a unit root for 9 of 11 apps in level. Only Zomato and Blinkit need differencing (d = 1).
- **Seasonal differencing.** STL seasonal strength is below 0.64 everywhere, so no app needs seasonal differencing (D = 0).

Weekday effect on 1-2★ reviews per app

Each day's count against its centred 7-day mean, averaged by weekday (feed-gap and source-gap days left out). Red = more complaints than usual on that weekday, blue = fewer.

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Swiggy	-13%	-13%	-5%	-7%	+6%	+5%	+28%
Zomato	-10%	-11%	-6%	-8%	+1%	+7%	+28%
Blinkit	-4%	-6%	-4%	-3%	-1%	-3%	+22%
Domino's	-14%	-10%	-14%	-10%	-5%	+14%	+38%
Myntra	-3%	+4%	-2%	+4%	-4%	-1%	-0%
Flipkart	-1%	+1%	-2%	-1%	-2%	+4%	+4%
Amazon	+1%	-0%	-3%	+1%	-4%	+3%	+1%
Meesho	-1%	+2%	-3%	-0%	-5%	+0%	+6%
Paytm	+1%	+0%	-2%	-2%	-2%	+4%	-2%
PhonePe	+3%	-2%	+2%	-2%	+6%	-2%	-9%
Google Pay	+11%	-1%	-0%	+11%	-6%	-11%	-8%

4.2 Implementation: ARIMA forecasting with a rolling-origin backtest

Models. Three models are fitted at every step:

Model	Description
ARIMA / SARIMA (main) [10]	Seasonal terms at lag 7 for the four food apps only. The feed-gap flag is an extra input for the five affected apps (SARIMAX). Gives 80% and 95% prediction intervals.
Seasonal naive (baseline)	The count on the same weekday last week.
Holt-Winters (comparison)	Damped additive trend with an additive weekly season.

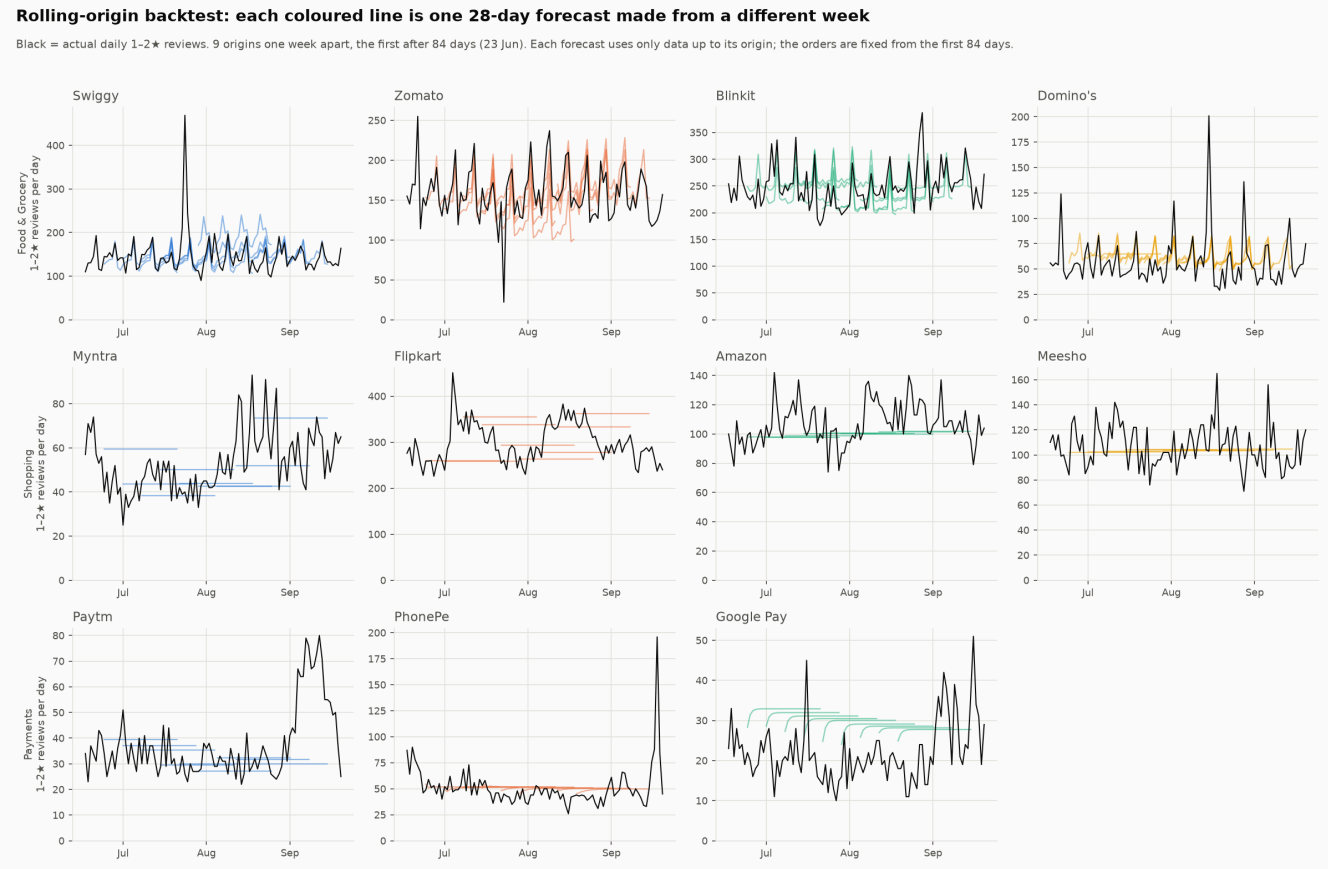
Choosing the orders:

1. d comes from the ADF test on the training days, and D = 0 for every app.
2. The ACF and PACF propose p and q up to 3, and a seasonal AR or MA term at lag 7 (always tried when the training days show a strong weekday effect).
3. The lowest AICc wins. If several orders are within 2 points of the lowest, the one with the fewest terms is taken. In total, 384 candidate fits were tried, and all converged.
4. There is no look-ahead: in the backtest, d, the weekday check and the orders are all chosen on the first 84 days only. On 84 days the ADF test asks for differencing for six apps, on all 173 days only for Zomato and Blinkit.

App	Final order $(p,d,q)(P,D,Q)_7$	Feed-gap input
Swiggy	(0,0,1)(1,0,1)	yes
Zomato	(2,1,1)(1,0,1)	
Blinkit	(1,1,1)(1,0,1)	yes
Domino's	(0,0,1)(1,0,1)	yes
Myntra	(3,0,2)	
Flipkart	(1,0,1)	yes
Amazon	(1,0,1)	yes
Meesho	(0,0,2)	
Paytm	(2,0,2)	
PhonePe	(1,0,0)	
Google Pay	(1,0,0)	

Only the food apps need seasonal terms, which matches their Sunday peak.

Rolling-origin backtest. Each model is trained on the first 84 days and forecasts the next 28. The origin then moves one week and every model is refitted. This gives 9 origins (23 June to 18 August) and 252 forecasts per app and model.



4.3 Evaluation

Accuracy. MAE and RMSE are in reviews per day. MASE divides each error by the seasonal-naive error on the training days of its own origin [11]. The relative MAE divides a model's MAE by the seasonal-naive MAE on the same backtest days, so below 1 means the model beats the baseline. Values are means over the 11 apps:

Model	MAE	RMSE	MASE	MAE ÷ seasonal-naive MAE	Lower MAE than seasonal naive
ARIMA	**19.39**	**26.74**	**0.93**	**0.93**	**10 of 11 apps**
Holt-Winters	19.14	26.66	0.93	0.88	10 of 11 apps
Seasonal naive	21.99	31.07	1.08	1.00	baseline

ARIMA's MAE per app, against the seasonal-naive baseline:

App	ARIMA	Seasonal naive	Holt-Winters
Swiggy	28.1	34.6	27.4
Zomato	22.8	23.9	21.5
Blinkit	28.8	33.8	26.8
Domino's	15.8	18.3	14.4
Myntra	12.2	13.9	12.8
Flipkart	52.7	56.0	54.7
Amazon	12.9	16.9	14.5
Meesho	13.6	19.3	15.1
Paytm	7.6	8.8	7.9
PhonePe	7.9	10.0	8.0

App	ARIMA	Seasonal naive	Holt-Winters
Google Pay	10.9	6.2	7.5

- **Where ARIMA gains most.** Meesho −29%, Amazon −24%, PhonePe −21% and Swiggy −19% against the baseline.
- **Google Pay is the exception.** Its level is low and noisy and it shifted after training, so "same day last week" adapts faster. Its MASE is still below 1 (0.64) because its training windows were noisier than the backtest days; the relative MAE (1.76) shows the baseline wins.
- **ARIMA and Holt-Winters are equally accurate.** ARIMA has the lower MAE for six apps (Amazon, Flipkart, Meesho, Myntra, Paytm, PhonePe), Holt-Winters for the four food apps; the means are 19.4 and 19.1. ARIMA stays the main model because it is the only one with prediction intervals, which the alerts and capacity planning need, and the only one that takes the feed gap as an input.

Prediction intervals. Across 2,772 backtest forecasts, 82.7% of days fell inside the 80% interval and 93.9% inside the 95% interval, close to nominal. By app, the 95% interval is too narrow for the shopping apps (Amazon 86%, Meesho 89%, Myntra 89%). Because rolling windows overlap, this is a descriptive check rather than an independent test.

Residuals. The Ljung–Box test at lag 14 [12] gives $p > 0.05$ for all 11 final models, so no autocorrelation is left at that lag. This does not prove that the residuals are random.

Unusual-day alerts. A forecast is flagged when the actual count falls outside its 95% interval:

- **Volume.** 169 forecasts were flagged. Because origins overlap, they cover 77 distinct app-days (56 above the interval, 21 below).
- **Apps with the most alerts.** Myntra has the most (15, ten of them below the interval), followed by Flipkart, Amazon and Meesho (11 each), Paytm (10) and Swiggy (9).
- **Largest alerts.** Swiggy on 24 July (469 against 136 expected, upper limit 177), Flipkart on 4 July (451 against 258) and Blinkit on 28 August (387 against 247).
- **What an alert means.** An alert marks an unexpected count. It does not prove a cause.

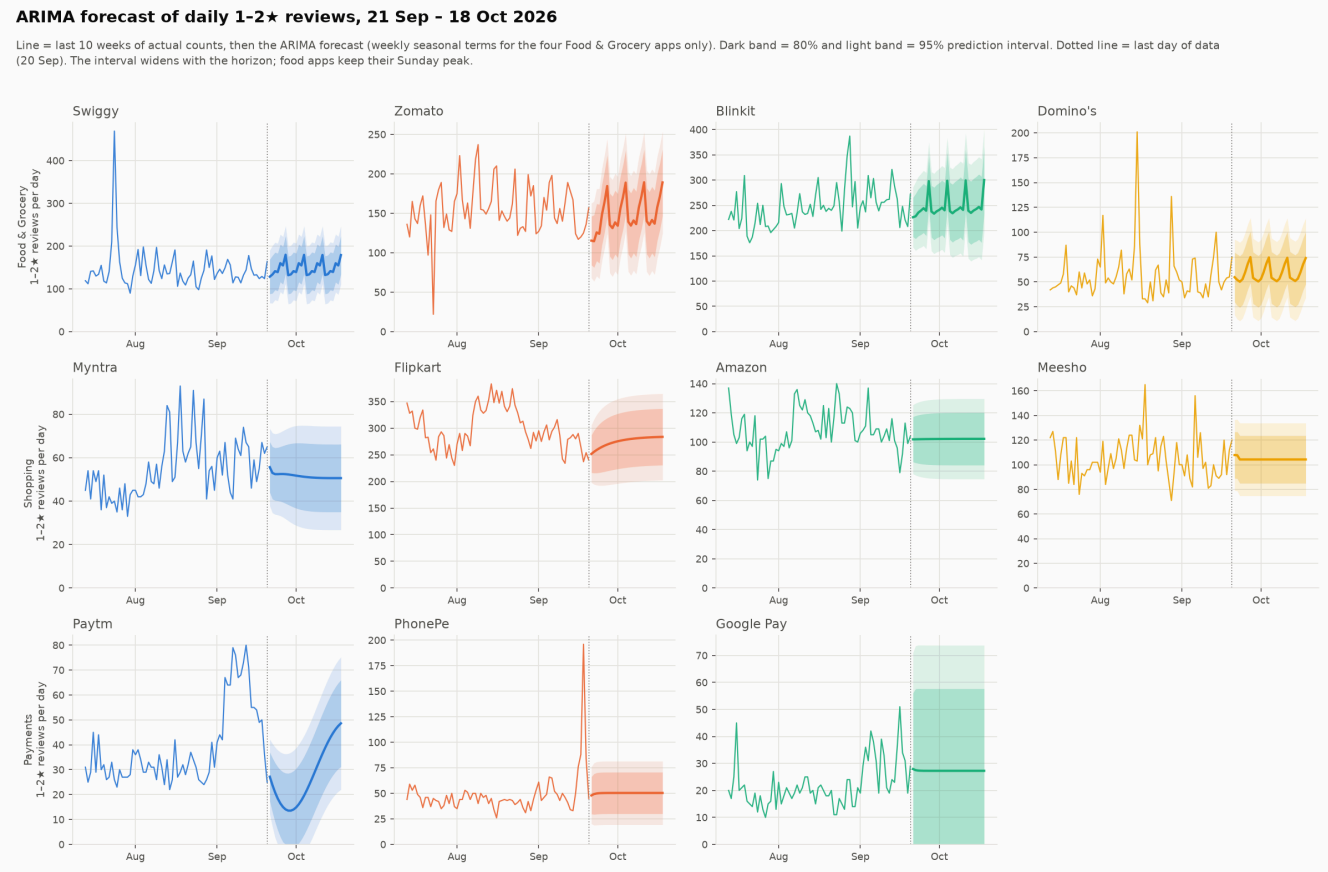
4.4 Interpretation: which apps can be predicted

Group	Apps	Evidence
Predictable	Zomato, Blinkit, Domino's	A steady weekly rhythm. The 95% interval holds 99% of days for Zomato, 97% for Blinkit and 95% for Domino's. Swiggy shares the rhythm but has short spikes (24 July), so its 95% interval holds 92%.
Hard to predict	Flipkart, Amazon, Meesho, Myntra	Level shifts. The 80% interval holds only 64–74% of days, and these apps have the most alerts.
Use with care	Google Pay, Paytm	Google Pay is low and noisy (the baseline wins). Paytm jumped to about 80 a day in September, so its week-1 forecast (−61%) is unreliable.

Forecast for 21–27 September (1–2★ reviews in the week after the data):

App	Forecast	80% interval	vs last 4 weeks
Flipkart	1,830	1,596 – 2,056	−7%
Blinkit	1,711	1,488 – 1,929	−8%
Swiggy	1,037	889 – 1,180	+7%
Zomato	977	811 – 1,151	−7%
Meesho	737	676 – 801	+4%

App	Forecast	80% interval	vs last 4 weeks
Amazon	714	642 – 786	−6%
Domino's	413	327 – 495	+7%
Myntra	371	318 – 424	−10%
PhonePe	348	270 – 431	−10%
Google Pay	192	105 – 310	+3%
Paytm	136	75 – 205	−61% (unreliable)



4.5 Update impact: did a new version make things worse?

Method. Each of the 224 releases is tested over 7-day windows. The release day is the day the version got its 10th review.

View	Comparison	Test
A. Before vs after (app level)	The app's 1–2★ reviews per day in the week before against the week from the release	Welch t-test
B. New vs older version (main test)	Over the same 7 days, the 1–2★ share of users on the new version against users still on older versions	Two-proportion z-test

View B is the main test because both groups cover the same days, so sales and outages affect both equally.

Usual gap. Users on the newest version are not a random sample. On Amazon and Flipkart they are always about 6 points more negative than users on older versions. Each release is therefore compared with its own app's usual gap (the median over the app's other releases).

Verdicts:

- **Worse:** at least 5 points beyond the usual gap, with q < 0.05 after Benjamini–Hochberg correction over all releases [13].
- **Confirmed at app level:** the app's 1–2★ reviews per day also rose significantly in View A.

Results.

Verdict	Releases
Worse, confirmed at app level	1
Worse, new version only	3
No clear change	196
Better	5
Not testable (window outside the data or too few reviews)	19
Total	**224**

App	Version	Release	1–2★ new vs older	Beyond usual gap	1–2★ per day before → after	Issue that rose most
Swiggy	4.107.1	19 Apr	60% vs 42%	+21.2 pts	162 → 150	Customer Support +8.9 pts
Swiggy	4.107.2	23 Apr	88% vs 83%	+8.5 pts	163 → 137	–
Flipkart	9.3	16 Apr	30% vs 16%	+7.6 pts	263 → 234	Customer Support +4.2 pts
Paytm	**10.84.5**	**3 Sep**	**23% vs 13%**	**+8.1 pts**	**34 → 66**	spread across issues

4 of 205 testable releases made things worse; most releases changed nothing measurable

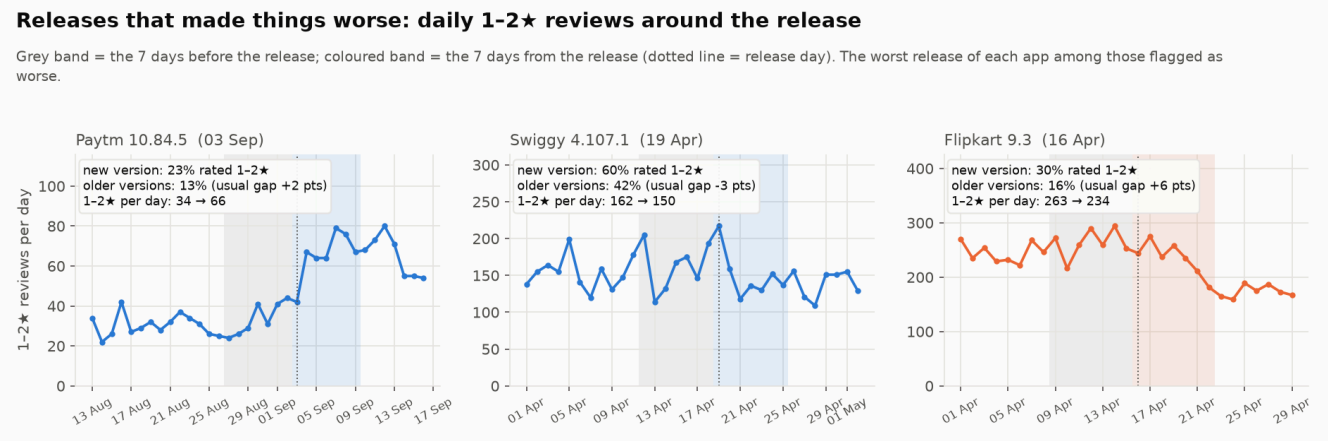
Verdict for each of the 224 release events. 'Worse' = over the same 7 days, the new version's share of 1–2★ reviews exceeds older versions' by at least 5 points more than the app's usual new-version gap (Benjamini–Hochberg q < 0.05). 'Confirmed' = the app's 1–2★ reviews per day also rose significantly against the week before.



Interpretation.

- **Paytm 10.84.5 is the clearest bad release.** Its users rated it lower than users on older versions (3.95 against 4.41), and the app's 1–2★ reviews per day nearly doubled. The forecast stage independently raised 6 alerts in the week after it. The release explains Paytm's September jump and its unreliable forecast.
- **The other three releases need care.** Their 7-day windows overlap the late-April feed gap, so their shares are inflated. The comparison is still fair, because both groups cover the same days.

- **The release that followed recovered.** Swiggy 4.108.0, which replaced the two worse Swiggy versions, is one of the 5 better releases.
- **Most apps had no bad release.** No release made things worse for Zomato, Blinkit, Domino's, Myntra, Amazon, Meesho, PhonePe or Google Pay.
- **The result is robust.** With the release day set to the first review on the version, the same 4 releases are flagged.



5. Combined Business Insights and Recommendations

Code: `scripts/22_combined_insights.py`. Tables are in `data/combined/` and the full write-up is `COMBINED_INSIGHTS.md`.

5.1 Priority rule

Each app scores one point per warning sign. A score of 3 or more is High priority, 2 is Medium and 0–1 is Low.

Warning sign	Rule	Source stage
Many 1–2★ per day	Above the median of the 11 apps (103 a day)	Time-series data
High 1–2★ share	Above the all-app share of 18.9% (feed-gap days excluded)	Time-series data
Rising trend	Last 4 weeks at least 10% above the first 4	Time-series patterns
Forecast above recent	A reliable week-1 forecast at least 5% above the last 4 weeks	Forecasting
Days above forecast	At least 5 days above the 95% interval	Forecast evaluation
Release made it worse	At least one worse release	Update impact

A forecast counts as **reliable** only if ARIMA beat the baseline in the backtest and the week-1 change is within ±30%. This excludes Paytm and Google Pay. The rule is deliberately simple: every point can be traced to one number in one stage.

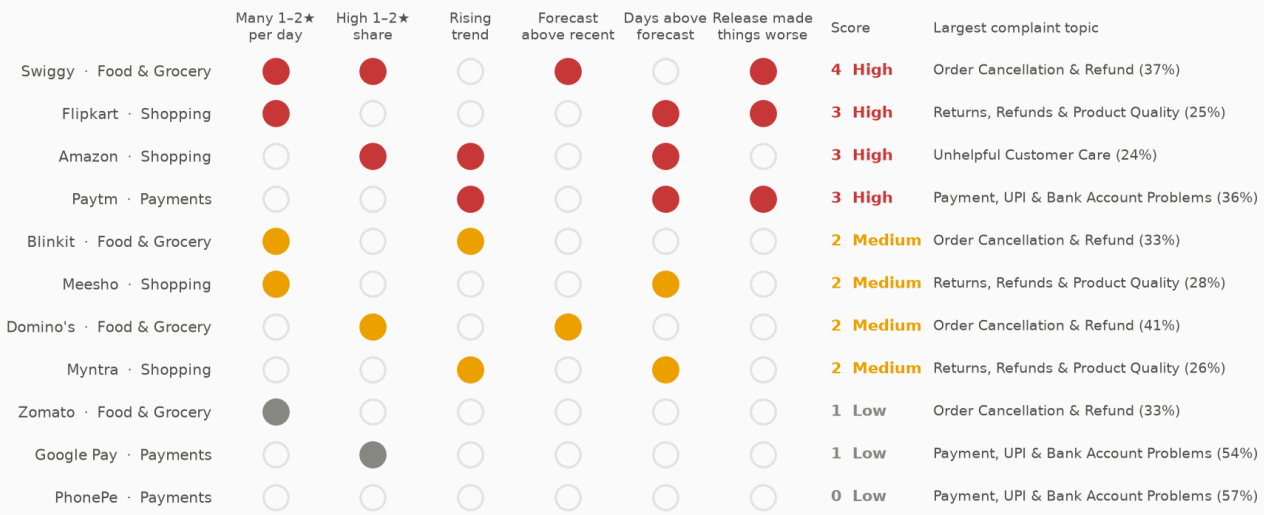
5.2 Priority table

App	Domain	Priority	1–2★ / day	1–2★ share	Trend	Largest complaint topic	Days above interval	Worse releases
Swiggy	Food & Grocery	**High (4)**	151	33.4%	−16%	Order Cancellation & Refund (37%)	4	2

App	Domain	Priority	1–2★ / day	1–2★ share	Trend	Largest complaint topic	Days above interval	Worse releases
Flipkart	Shopping	**High (3)**	282	17.7%	+5%	Returns, Refunds & Product Quality (25%)	11	1
Amazon	Shopping	**High (3)**	103	53.4%	+14%	Unhelpful Customer Care (24%)	9	0
Paytm	Payments	**High (3)**	38	14.6%	+60%	Payment, UPI & Bank Account Problems (36%)	10	1 (confirmed)
Blinkit	Food & Grocery	Medium (2)	251	18.4%	+13%	Order Cancellation & Refund (33%)	3	0
Meesho	Shopping	Medium (2)	104	17.6%	+2%	Returns, Refunds & Product Quality (28%)	9	0
Domino's	Food & Grocery	Medium (2)	60	22.6%	−18%	Order Cancellation & Refund (41%)	4	0
Myntra	Shopping	Medium (2)	51	10.3%	+22%	Returns, Refunds & Product Quality (26%)	5	0
Zomato	Food & Grocery	Low (1)	164	18.8%	−11%	Order Cancellation & Refund (33%)	0	0
Google Pay	Payments	Low (1)	27	23.2%	−20%	Payment, UPI & Bank Account Problems (54%)	0	0
PhonePe	Payments	Low (0)	50	9.8%	+3%	Payment, UPI & Bank Account Problems (57%)	1	0

Which apps need attention first: six warning signs from all stages

Filled dot = the sign applies. Score = number of signs (3+ High, 2 Medium, 0–1 Low). Signs: 1–2★ reviews per day above the median app; share of 1–2★ reviews above all apps; last 4 weeks 10%+ above the first 4; reliable week-1 forecast 5%+ above the last 4 weeks; 5+ days above the 95% forecast interval; a release that made things worse.



5.3 Recommendations per domain

Domain	Fix first (most common complaint)	Most damaging (angriest specific complaint)	Focus
Food & Grocery	**Refund and cancellation flow**: 35% of the domain's complaints (28,653), the largest topic in every food app	**Customer support**: −0.40, 56% strongly negative	Swiggy (High). Blinkit has the most complaints, 251 a day
Shopping	**Returns, refunds and product quality**: 25% (18,123), the largest topic in Flipkart, Meesho and Myntra	**Orders cancelled or not delivered**: −0.31, 41% strongly negative	Flipkart and Amazon (High)
Payments	**Payment, UPI and bank-account problems**: 49% (6,443)	**Unanswered support**: −0.23, angrier than the payment problem itself (−0.15)	Paytm (High). PhonePe has the most complaints, 50 a day, but only 9.8% of its reviews

"Most damaging" is the most negative topic that names a fault; the general "bad" and "worst ever" topics are angrier but say only that users are unhappy.

5.4 Recommendations for the High-priority apps

Swiggy (4 signs)

- Fix refunds and cancellations first: they are 37% of its complaints, with a sentiment of −0.33.
- Review version 4.107.1. Its users were 21 points more negative than Swiggy's usual gap, and support complaints rose 8.9 points.
- Plan support for about 1,037 1–2★ reviews in the week of 21–27 September (80% interval 889–1,180).

Flipkart (3 signs)

- Fix returns, refunds and product quality first: 25% of its complaints.
- Set a daily alert. Flipkart had 11 days above the forecast interval, the most of any app; the largest was 451 against 258 expected on 4 July.
- Review version 9.3, after which support and return complaints rose.

Amazon (3 signs)

- Fix customer care first: unhelpful customer care is 24% of its complaints.
- Amazon has the highest 1–2★ share of all 11 apps (53%), and it is rising (+14%).
- Use the Review 1 classifier to triage reviews: it is most useful here, with 91% precision and 96% recall.

Paytm (3 signs)

- Investigate version 10.84.5 first. It is the only release confirmed at the app level, and it took 1–2★ reviews from 34 to 66 a day.
- Pay out promised cashback and rewards: it is Paytm's angriest specific complaint (18% of its complaints, −0.23).
- Plan capacity on last week's level, not on the ARIMA forecast, which is unreliable after the September jump.
- Have a person check flagged reviews, because the classifier's precision is lowest here (66%).

The Medium- and Low-priority apps get the same four kinds of recommendation: fix first, most damaging, plan capacity and triage. In total there are 65 recommendations (`data/combined/recommendations.csv`), and all of them are shown on the dashboard.

---

6. Interactive Dashboard

The dashboard is a Streamlit app (`dashboard/`), deployed on Streamlit Community Cloud at **https://team10-app-reviews.streamlit.app/**. It redeploys from the `master` branch after every merge. It reads only small precomputed tables, so it opens in seconds.

Every page shares the same sidebar filters: domain, app, date range, and an option to exclude the feed gap.

Page	What the user can do	Built by
Overview	Headline numbers, 1–2★ reviews per day by domain (four measures, 7-day smoothing), share and count of 1–2★ reviews by app, top keyword issues, table with CSV download	Aditya
Text mining	Topic explorer with top terms, sentiment and real example reviews (189 readable examples); keyword-tag overlap; topic × app heatmap; topic share over time; distinctive terms per app	Akshay
Forecasting	ARIMA forecast per app with 80% and 95% intervals, optional baseline comparison, unusual days above and below the interval, release markers (red = made things worse), weekly forecast table and backtest error relative to the seasonal-naive baseline	Kanishka
Update impact	Verdict for every release, new-vs-older gap against the app's usual gap, daily series with releases, before/after detail and issue mix of any release, CSV download	Kanishka
Recommendations	Priority counts, warning-sign grid, priority table and recommendations, filtered by domain and app, with downloads	Kanishka
Check a review	Paste any review: the Review 1 classifiers' problem probability with an adjustable threshold, the complaint topic from the saved NMF model, VADER scores and keyword tags	Harshini

Deploy

Filters

Domain

Food & Grocery × Shopping × Payments ×

App

Swiggy × Zomato × Blinkit × Domino's × Myntra × Flipkart × Amazon × Meesho × Paytm × PhonePe ×

Date range

1 Apr — 20 Sep

Exclude the 21 Apr – 5 May feed gap

Data: every English/India Play Store review of 11 apps, 1 Apr – 20 Sep 2026.

Overview

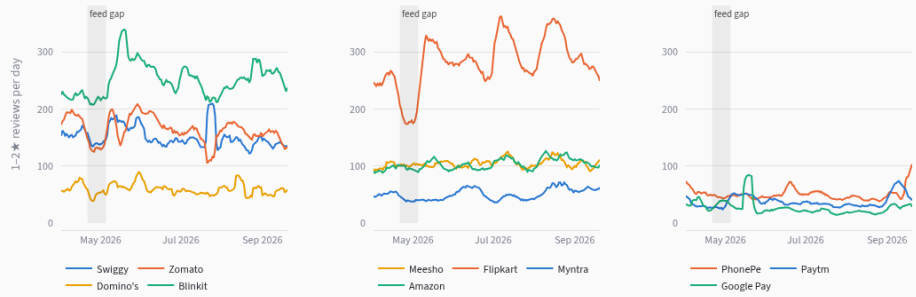
11 app(s) · 01 Apr – 20 Sep 2026

Reviews	1–2★ reviews	Share rated 1–2★	Average rating	1–2★ reviews per day
1,137,987	221,581	19.5%	4.06★	1,281

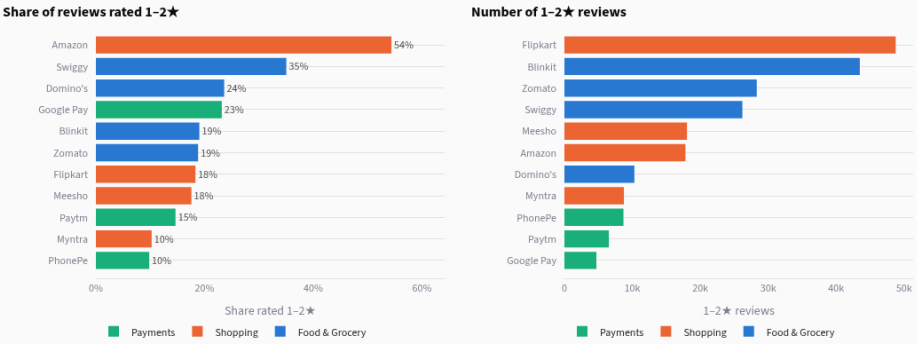
Over time

Measure: 1–2★ reviews per day | Share rated 1–2★ | Average rating | All reviews per day

7-day average



By app



Top issues per app

	Customer Support	Cancellation & Return	Delivery Delay	Payment & Refund	Pricing & Fraud	Crash & Stability	Order Quality	UI/UX & Update	Account / Login / OTP
Swiggy	9.9	5.3	6.1	4.1	3.5	0.6	1.9	0.2	0.2
Zomato	4.5	1.5	3.2	1.7	2.0	0.2	1.0	0.1	0.0
Blinkit	3.0	2.3	2.9	1.5	1.5	0.3	1.2	0.1	0.0
Domino's	3.9	2.0	3.3	1.1	1.0	0.5	0.7	0.1	0.1
Myntra	3.7	5.0	2.2	1.5	1.3	0.3	0.9	0.3	0.1
Flipkart	3.4	3.2	2.6	1.0	1.4	0.4	0.6	0.1	0.2
Amazon	16.8	12.5	8.9	7.4	4.2	2.0	2.8	0.8	0.8
Meesho	3.8	5.5	2.6	2.0	1.7	0.5	0.9	0.1	0.5
Paytm	1.8	0.4	0.1	0.6	1.1	2.2	0.0	0.5	0.5
PhonePe	0.9	0.1	0.1	0.4	0.6	0.5	0.0	0.2	0.2
Google Pay	1.6	0.4	0.1	1.1	1.3	2.7	0.0	0.7	0.4

% of reviews: 15, 10, 5

% of each app's reviews that mention the issue (keyword tags; a review can have several). Most reviews are too short to name an issue, so these are lower bounds.

› Table by app

7. Limitations

- **Short window.** Only 173 days of data are available, so there is no yearly pattern and the backtest has only 9 origins.
- **Counts depend on things the models do not see.** Review counts follow app traffic and campaigns. The forecast answers "what if the next weeks look like the recent past".
- **A 1–2★ review is a judgement, not a confirmed defect.** It is the user's own judgement. The issue mix in the update-impact test uses keyword tags, which miss complaints worded differently.
- **Topic names are our reading of the top terms.** Three topics are general dissatisfaction ("bad", "not good", "worst ever") rather than one fault; they are named so and are not used to pick what to fix. The example reviews on the dashboard let a reader check each name.
- **Keyword tags are broad.** The 9 keyword tags (Review 1) match some praise too: 11.6% of all tags fall on 4–5★ reviews (for example "delivery boy was good" counts as a delivery mention). Text mining therefore uses only 1–2★ reviews, and its topics do not depend on the tags.
- **Forecast intervals are narrow for shopping apps.** The 95% interval holds 86–89% of days for Amazon, Meesho and Myntra, so these apps raise more alerts than a well-calibrated model would.
- **VADER underrates calm technical complaints.** Sentiment shows how users feel, not how severe a fault is.
- **Users choose when to update.** Early updaters may differ from late ones, which is why each release is compared with its app's usual gap. Three of the four worse releases have windows that overlap the feed gap.
- **The priority score is simple.** It weights every sign equally and uses fixed thresholds. Each sign is shown separately, so the reason for a priority is always visible.

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8. Project Tracking and Individual Contributions

Board. The GitHub Projects board (https://github.com/users/Adithya-Monish-Kumar-K/projects/4) uses the stages Backlog → To Do → In Progress → Review/Testing → Completed.

- **Tasks.** Review 2 has 14 tasks, each assigned to one member with a due date of 11 October. All 14 are Completed.
- **Pull requests.** Every task is linked to a merged pull request, and every issue is closed.
- **Order of work.** The work was done as a relay: each stage started on the merged work of the stage before.

	Title	Assignees	Status	Linked pull requests	Due Date
1	Build daily and weekly time series for forecasting #131	Adithya-Monish-K...	Completed	#147	Oct 11, 2026
2	Preprocess 1–2 star review text for text mining #132	Adithya-Monish-K...	Completed	#148	Oct 11, 2026
3	Build dashboard skeleton, filters and overview page #133	Adithya-Monish-K...	Completed	#149	Oct 11, 2026
4	Topic modelling and distinctive terms per app #134	RKSAKETH	Completed	#150	Oct 11, 2026
5	Build forecasting models with rolling backtest #135	akshayks13	Completed	#152	Oct 11, 2026
6	Evaluate and interpret the topic models #136	RKSAKETH	Completed	#151	Oct 11, 2026
7	Evaluate forecasts and flag unusual days #137	Harshini3105	Completed	#154	Oct 11, 2026
8	Analyse update impact before and after each app version #138	d-kanishka	Completed	#157	Oct 11, 2026
9	Dashboard: text-mining page #139	akshayks13	Completed	#153	Oct 11, 2026
10	Dashboard: forecasting, update-impact and recommendations pages #140	d-kanishka	Completed	#158	Oct 11, 2026
11	Write combined business insights and recommendations #141	d-kanishka	Completed	#157	Oct 11, 2026
12	Write up forecast results #142	Harshini3105	Completed	#155	Oct 11, 2026
13	Assemble final Jupyter notebook #143	d-kanishka	Completed	#158	Oct 11, 2026
14	Dashboard: check-a-review page #146	Harshini3105	Completed	#156	Oct 11, 2026

Member	Register number	Stage	Board tasks	Pull requests	Evidence
Aditya Monish Kumar K	CB.SC.U4CSE23103	1 and 6	#131 daily and weekly series, release events and patterns; #132 text preprocessing; #133 dashboard skeleton, filters, overview page and deployment. Stage 6: slide deck, final report, contribution summary	#147, #148, #149	`scripts/15_time_series_data.py`, `scripts/16_text_preprocessing.py`, `data/timeseries/`, `data/textmining/`, `dashboard/app.py`, `dashboard/common.py`, `dashboard/pages/overview.py`, `FINAL_REPORT.md`
Regella Krishna Saketh	CB.SC.U4CSE23649	2	#134 topic modelling, distinctive terms and sentiment per topic; #136 coherence, keyword-tag comparison, untagged reviews and topic trends	#150, #151	`scripts/17_topic_modelling.py`, `scripts/18_topic_evaluation.py`, `models/textmining/`, `TEXT_MINING.md` §2–3
Akshay KS	CB.SC.U4CSE23104	3	#135 ARIMA/SARIMAX forecasting with rolling backtest; #139 text-mining dashboard page	#152, #153	`scripts/19_forecasting.py`, `scripts/20_topic_examples.py`, `dashboard/pages/text_mining.py`, `TIME_SERIES.md` §2.1–2.4, 2.6–2.8
Harshini Vennela	CB.SC.U4CSE23455	4	#137 forecast evaluation, residual checks and unusual-day alerts; #142 forecast write-up; #146 check-a-review page	#154, #155, #156	`scripts/20_forecast_evaluation.py`, `dashboard/pages/check_review.py`, `TIME_SERIES.md` §2.5
Kanishka D	CB.SC.U4CSE23155	5	#138 update impact; #141 combined insights and recommendations; #140 forecasting, update-impact and recommendations pages; #143 final notebook	#157, #158	`scripts/21_update_impact.py`, `scripts/22_combined_insights.py`, `dashboard/pages/forecasting.py`, `update_impact.py`, `recommendations.py`, `COMBINED_INSIGHTS.md`, `notebooks/Review_2.ipynb`

The detailed contribution summary is `docs/Review_2_Contribution_Summary.md`.

9. Conclusion

- **Users complain mostly about service, not about the app software.** Refunds and cancellations, returns, payment problems and support lead every domain. Among specific faults, unanswered support draws the angriest reviews.
- **Text mining finds what keyword rules miss.** NMF beats LDA on held-out coherence in every domain. The topics agree with the keyword tags where both apply, and they give a theme to the 43% of complaints that no keyword matched, including cash on delivery and unpaid cashback, which have no keyword category.
- **Daily complaints can be forecast well enough to plan support.** ARIMA beats the baseline for 10 of 11 apps, is as accurate as Holt-Winters, and adds intervals that hold 94% of days at the 95% level. Its alerts mark days that need a look. Shopping apps with level shifts, Google Pay and Paytm need more care.
- **Most app updates do not hurt.** Of 205 testable releases, 196 changed nothing measurable. Paytm 10.84.5 clearly made things worse, and the forecast caught it independently.
- **Fix first:** Swiggy, Flipkart, Amazon and Paytm. The live dashboard puts all of this, and a check for any single review, in front of a product or support team.

Together, Reviews 1 and 2 turn 1.1 million unstructured reviews into a monitoring tool. A team can see what is going wrong, whether it is getting worse, whether an update caused it, and what to fix first.

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10. How to Reproduce (Review 2)

```
pip install -r requirements.txt
python -c "import nltk; nltk.download('vader_lexicon')"

python scripts/15_time_series_data.py      # daily/weekly series, release events, STL, ADF,
ACF/PACF
python scripts/16_text_preprocessing.py    # complaint and praise corpora, document-term
matrices
python scripts/17_topic_modelling.py       # NMF topics, distinctive terms, sentiment per
topic
python scripts/18_topic_evaluation.py      # coherence, keyword-tag overlap, untagged reviews,
trends
python scripts/20_topic_examples.py        # readable example reviews for the dashboard
python scripts/19_forecasting.py           # ARIMA/SARIMAX, baselines, rolling backtest, 4-
week forecast
python scripts/20_forecast_evaluation.py   # MAE/RMSE/MASE, interval coverage, Ljung-Box,
alerts
python scripts/21_update_impact.py         # before/after and new-vs-older tests per release
python scripts/22_combined_insights.py     # priority table and recommendations
python scripts/23_topic_model_comparison.py   # NMF vs LDA on a 20% held-out split (~2 min)
python scripts/24_dataset_documentation.py    # data dictionary, sample, validation summary
python scripts/25_train_only_preprocessing_check.py  # Unit 2 with preprocessing fitted on
training rows (~15 min)

streamlit run dashboard/app.py             # the dashboard
```

The Review 1 steps (scripts 01–14) are listed in `REPORT.md` §11.

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11. References

[1] D. Pagano and W. Maalej, "User feedback in the AppStore: An empirical study," IEEE International Requirements Engineering Conference (RE), 2013.

[2] N. Chen, J. Lin, S. C. H. Hoi, X. Xiao and B. Zhang, "AR-Miner: Mining informative reviews for developers from mobile app marketplace," International Conference on Software Engineering (ICSE), 2014.

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